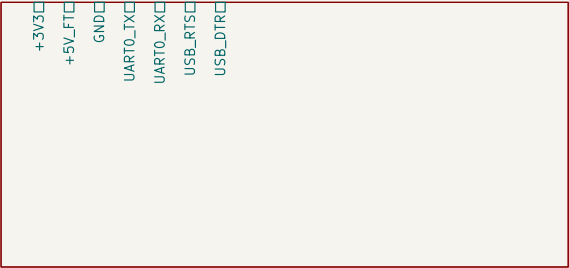


A3 CONTROL CENTER – ESP32 DUAL-CAN / OLED / COMPOSITE VIDEO

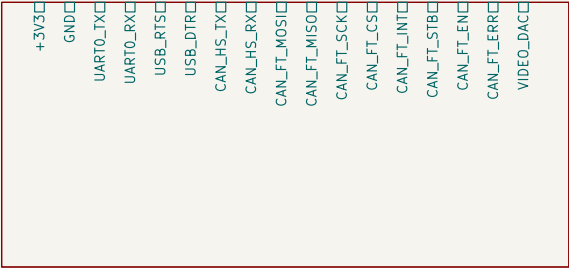
Revision A schematic capture – reviewed hierarchical nets connect the functional sheets.

USB +3V3 GND UART0_TX UART0_RX USB_DTR



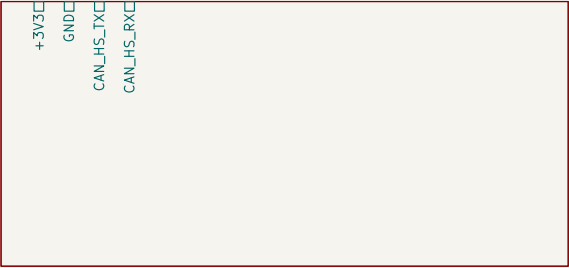
File: dual_can_esp32_power_usb.kicad_sch

ESP32 +3V3 GND UART0_TX UART0_RX USB_RTSC USB_DTR CAN_HS_TX CAN_HS_RX CAN_FT_MOSI CAN_FT_MISO CAN_FT_SCK CAN_FT_CS CAN_FT_INT CAN_FT_STB CAN_FT_ENC CAN_FT_ERR VIDEO_DAC



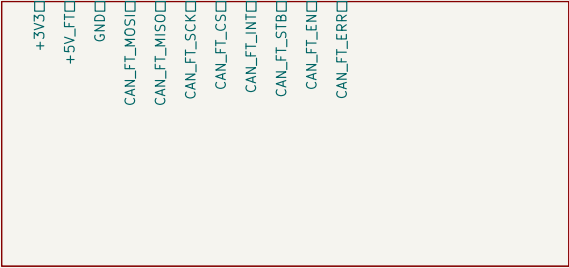
File: dual_can_esp32_mcu_ui.kicad_sch

CAN_HS GND CAN_HS_TX CAN_HS_RX



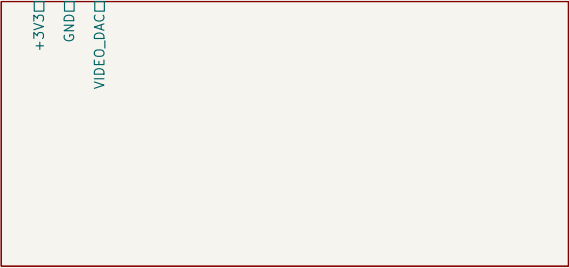
File: dual_can_esp32_can_hs.kicad_sch

CAN_FT +3V3 GND CAN_FT_MOSI CAN_FT_MISO CAN_FT_SCK CAN_FT_CS CAN_FT_INT CAN_FT_STB CAN_FT_ENC CAN_FT_ERR



File: dual_can_esp32_can_ft.kicad_sch

Composite Video +3V3 GND VIDEO_DAC



File: dual_can_esp32_video.kicad_sch

Sheet: /
File: dual_can_esp32.kicad_sch

Title: dual_can_esp32

Size: A4

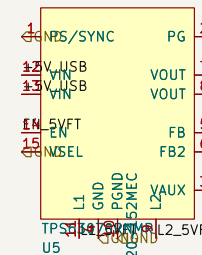
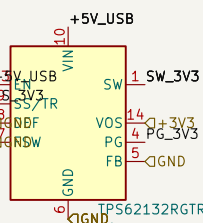
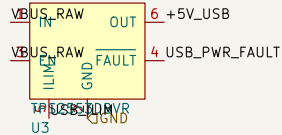
Date:

KiCad E.D.A. 10.0.6

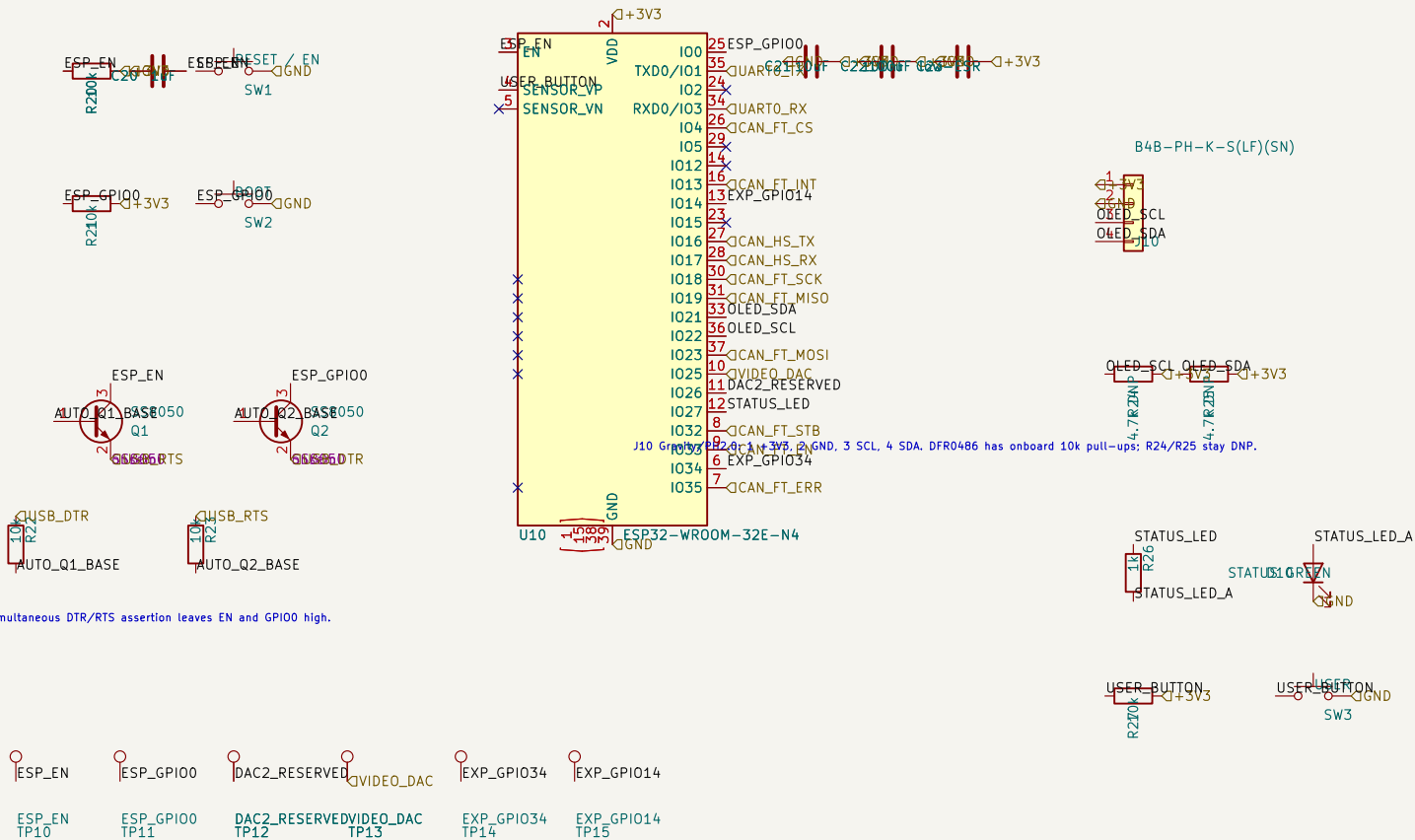
Rev:

Id: 1/6

Pin	Signal	Function
21	CP	26 QUART0_RX
22	RS1	25 QUART0_TX
23	VREG_RTS	24 USB_RTS
24	VD0	23 CTS
25		22 DSR
26		21 DTR
27		20 USB_DTR
28		19 DCD
29		18 RI/CLK
30		17 SUSPEND
31		16 SUSPEND
32		15 TX0/GPIO.0
33		14 RX1/GPIO.1
34		13 RS485/GPIO.2
35		12 WAKEUP/GPIO.3
36		11 GPIO.4
37		10 GPIO.5
38		9 GPIO.6
39		8 CHREN
40		7 CHRO



ESP32-WROOM-32E, PROGRAMMING, DISPLAY AND USER I/O



NPN auto-program circuit: simultaneous DTR/RTS assertion leaves EN and GPIO0 high.

Sheet: /ESP32 + UI/
File: dual_can_esp32_mcu_ui.kicad_sch

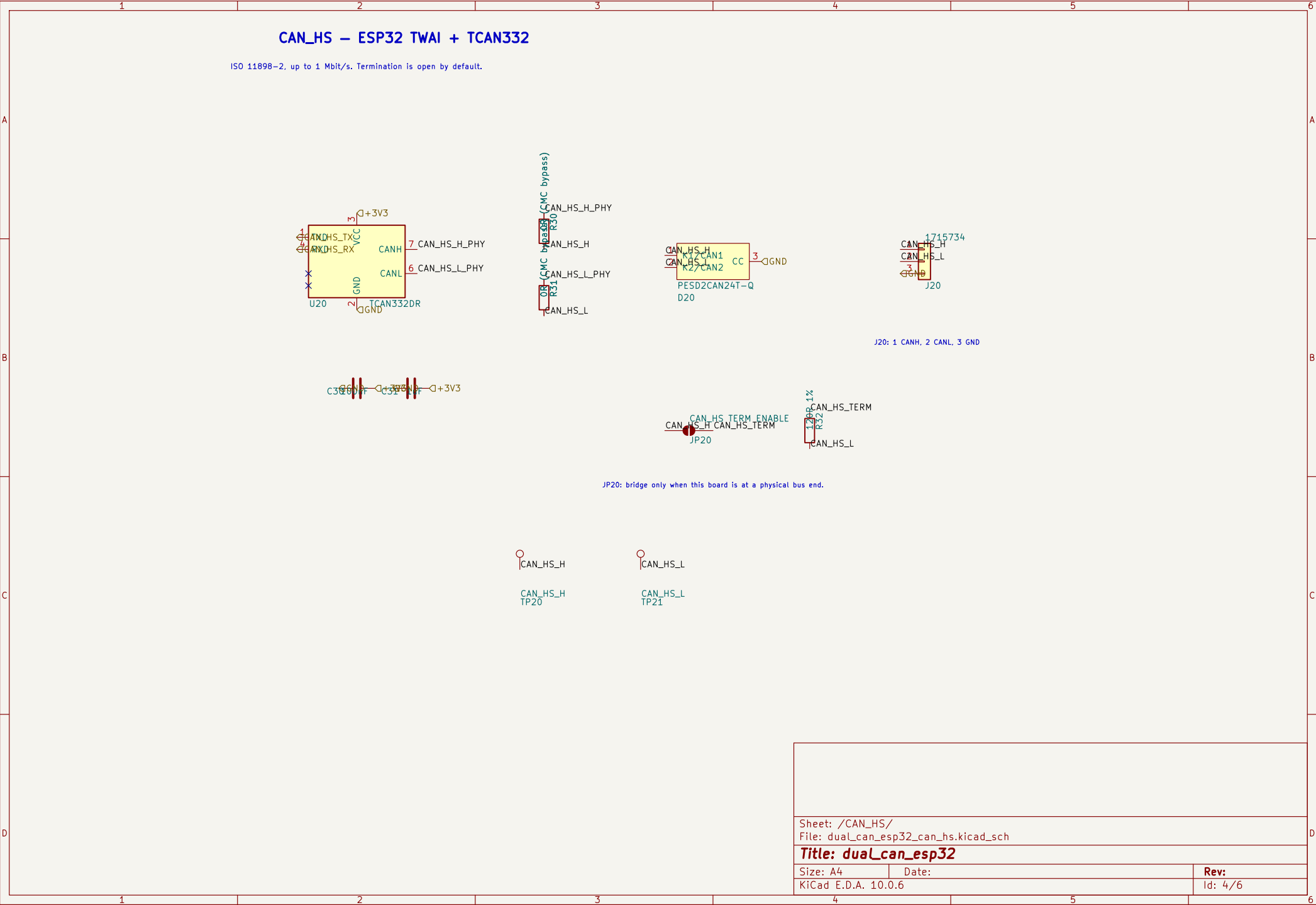
Title: dual_can_esp32

Size: A4 Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 3/6

[illegible][illegible]

CAN_HS – ESP32 TWAI + TCAN332

ISO 11898–2, up to 1 Mbit/s. Termination is open by default.

U20: TCAN332DR

U20: PESD2CAN24T-Q D20

JP20: bridge only when this board is at a physical bus end.

J20: 1 CANH, 2 CANL, 3 GND

Sheet: /CAN_HS/
File: dual_can_esp32_can_hs.kicad_sch
Title: dual_can_esp32
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.6 Id: 4/6

CAN_HS – ESP32 TWAI + TCAN332

ISO 11898–2, up to 1 Mbit/s. Termination is open by default.

U20: TCAN332DR

U20: PESD2CAN24T-Q D20

JP20: bridge only when this board is at a physical bus end.

J20: 1 CANH, 2 CANL, 3 GND

Sheet: /CAN_HS/
File: dual_can_esp32_can_hs.kicad_sch
Title: dual_can_esp32
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.6 Id: 4/6

CAN_HS – ESP32 TWAI + TCAN332

ISO 11898–2, up to 1 Mbit/s. Termination is open by default.

U20: TCAN332DR

U20: PESD2CAN24T-Q D20

JP20: bridge only when this board is at a physical bus end.

J20: 1 CANH, 2 CANL, 3 GND

Sheet: /CAN_HS/
File: dual_can_esp32_can_hs.kicad_sch
Title: dual_can_esp32
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.6 Id: 4/6

ISO 11898-3, up to 125 kbit/s. JP40/JP41 are open by default; fit both together only after network termination review.

R44/R45 force STB=0, EN=0 (non-driving standby) during ESP32 reset; firmware sets both high for normal mode.

1kpf loads assume about 3 pF total stray. Validate frequency/startup on assembled PCB.

Sheet: /CAN_FT/		
File: dual_can_esp32_can_ft.kicad_sch		
Title: dual_can_esp32		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 5/6

CL=12 pF; 18 pF loads assume about 3 pF total stray. Validate frequency/startup on assembled PCB.

COMPOSITE VIDEO - GPIO25 DAC + THS7314

Default: 0.1 uF AC input clamp, THS7314 CH1, 75 ohm source, DC output. Bench--calibrate DAC levels before freeze.

J60: 1 RCA center VIDEO_OUT, 2 shell VIDEO_GND. Exact THT footprint remains a mechanical gate.

Unused CH2/CH3 inputs tied to GND; outputs left NC.

Interlock: only one of JP60, JP61, or JP62 may be closed. JP60 is the assembly default.

Sheet: /Composite Video/
File: dual_can_esp32_video.kicad_sch
Title: dual_can_esp32
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.6 Id: 6/6

Title: <u>dua_Lcan_esp32</u>		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 6/6

Size: A4	Date:	Rev:
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Rev:

Id: 6/6